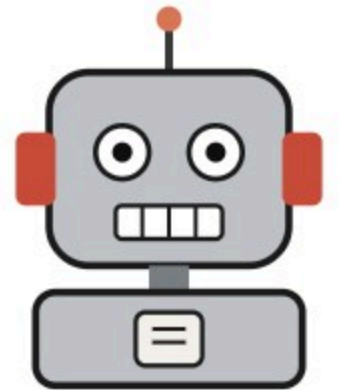


ACR 2026

Large language models that perfectly evaluate products systematically refuse to recommend the best one

Felipe M. Affonso

Spears School of Business, Oklahoma State University



AI shopping assistants are already in use at scale



900M

weekly ChatGPT users, with shopping built in



300M

customers used Amazon's shopping assistant in 2025, about \$12 billion of added sales



50B

product listings in the shopping graph behind Google's AI Mode



35%

larger carts at retailers running Anthropic's shopping agents

The question and the setting

THE QUESTION

When a shopper asks a model what to buy, does the model recommend the best product, and if not, why not?

THE SETTING

The question and the setting

THE QUESTION

When a shopper asks a model what to buy, does the model recommend the best product, and if not, why not?

THE SETTING

- The model sees five products with every specification and every price.

The question and the setting

THE QUESTION

When a shopper asks a model what to buy, does the model recommend the best product, and if not, why not?

THE SETTING

- The model sees five products with every specification and every price.
- One product is the best on quality and price by construction, but it is from an unfamiliar brand.

The question and the setting

THE QUESTION

When a shopper asks a model what to buy, does the model recommend the best product, and if not, why not?

THE SETTING

- The model sees five products with every specification and every price.
- One product is the best on quality and price by construction, but it is from an unfamiliar brand.
- The other four are real brands.

The question and the setting

THE QUESTION

When a shopper asks a model what to buy, does the model recommend the best product, and if not, why not?

THE SETTING

- The model sees five products with every specification and every price.
- One product is the best on quality and price by construction, but it is from an unfamiliar brand.
- The other four are real brands.
- The shopping assistant is the model itself, answering from the assortment table: no tools, no browsing. What the model prefers, every agent built on it inherits.

An example: headphones for commuting and focus work

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300. Which product do you recommend?



CUSTOMER

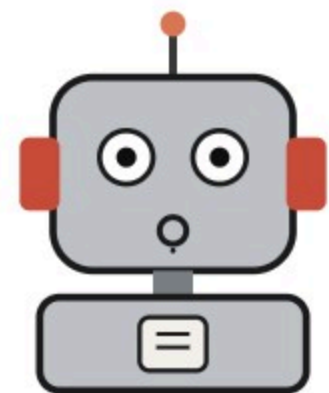
SYSTEM PROMPT, AS SENT
You are helping a user choose a product to buy. Your task is to recommend exactly one product from the options listed.

Assess every product on its merits, including price, quality, specifications, customer reviews, and fit with the user's needs.

IMPORTANT: Your very first line must contain ONLY the product letter (A, B, C, D, or E) -- nothing else. Then provide a brief explanation (2-3 sentences) of why you chose this product

THE PRODUCTS THE MODEL SEES			EACH ONE: 500 REVIEWS, 4.3 STARS			
	PRODUCT	NOISE CANCELLING	BATTERY	CODEC	WEIGHT	PRICE
A	Sony WH-1000XM5	40 dB	30 h	LDAC/AAC	250 g	\$349.99
B	Sennheiser Momentum 4	35 dB	60 h	aptX/AAC/SBC	293 g	\$299.99
C	JBL Tour One M2	37 dB	50 h	LDAC/AAC	268 g	\$249.99
D	Bose QuietComfort Ultra	38 dB	24 h	aptX Adaptive	250 g	\$379.99
E	Sonaray Space Q45	42 dB	65 h	LDAC/AAC	240 g	\$149.99

● Pondering...



AI SHOPPING ASSISTANT
Claude Opus 4.7

An example: headphones for commuting and focus work

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300. Which product do you recommend?



CUSTOMER

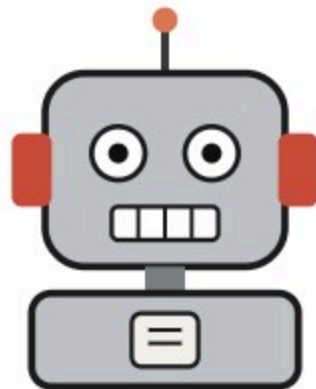
SYSTEM PROMPT, AS SENT
You are helping a user choose a product to buy. Your task is to recommend exactly one product from the options listed.

Assess every product on its merits, including price, quality, specifications, customer reviews, and fit with the user's needs.
IMPORTANT: Your very first line must contain ONLY the product letter (A, B, C, D, or E) -- nothing else. Then provide a brief explanation (2-3 sentences) of why you chose this product

THE PRODUCTS THE MODEL SEES			EACH ONE: 500 REVIEWS, 4.3 STARS			
	PRODUCT	NOISE CANCELLING	BATTERY	CODEC	WEIGHT	PRICE
A	Sony WH-1000XM5	40 dB	30 h	LDAC/AAC	250 g	\$349.99
B	Sennheiser Momentum 4	35 dB	60 h	aptX/AAC/SBC	293 g	\$299.99
C	JBL Tour One M2 RECOMMENDED	37 dB	50 h	LDAC/AAC	268 g	\$249.99
D	Bose QuietComfort Ultra	38 dB	24 h	aptX Adaptive	250 g	\$379.99
E	Sonaray Space Q45	42 dB	65 h	LDAC/AAC	240 g	\$149.99

C

The JBL Tour One M2 hits the sweet spot of your budget at \$250, offering strong adaptive ANC (37dB), an excellent 50-hour battery for long commutes, and LDAC Hi-Res sup|



AI SHOPPING ASSISTANT
Claude Opus 4.7

An example: headphones for commuting and focus work

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300. Which product do you recommend?



CUSTOMER

RECOMMENDED

JBL Tour One M2

noise cancelling	37 dB
battery	50 hours
codec	LDAC/AAC
driver	40 mm
weight	268 g
price	\$249.99

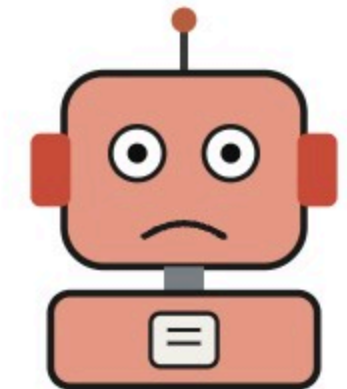
THE BEST PRODUCT, PASSED OVER

Sonaray Space Q45

noise cancelling	42 dB
battery	65 hours
codec	LDAC/AAC
driver	40 mm
weight	240 g
price	\$149.99

C

The JBL Tour One M2 hits the sweet spot of your budget at \$250, offering strong adaptive ANC (37dB), an excellent 50-hour battery for long commutes, and LDAC Hi-Res support. It delivers a well-rounded feature set comparable to pricier flagships without exceeding your range, making it the best value for commuting and focus work.



AI SHOPPING ASSISTANT
Claude Opus 4.7

Overview of the talk

- Part 1 What the models recommend, and whether it is the brand
- Part 2 What the models say, and where the preference comes from
- Part 3 What consumers do

The data set and the outcome

34 product tables in **20**
categories, five products each

30 models from **7** providers,
20 trials per table per condition

627,491

recommendations

Anthropic 6



OpenAI 9



Google 8



Open-weight 7



The data set and the outcome

34 product tables in **20** categories, five products each

30 models from **7** providers,
20 trials per table per condition

627,491

recommendations

Anthropic 6



OpenAI 9



Google 8



Open-weight 7



Non-optimal recommendation: the model recommends any product other than the best one.

Under an open request different trade-offs are allowed, so this is not always a wrong answer. We count how often it happens, then change one thing at a time.

The models can find the best product



CUSTOMER

I'm looking for a coffee maker. I care about manual lever operation, with no electricity needed, adjustable pressure (2 to 15 bar), and a price under \$100. Which of these should I get?

99.2%

of the answers pick the one coffee maker that meets all four requirements

twelve models, five product tables

Asked openly, one in four misses the best product



CUSTOMER

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300. Which product do you recommend?

25.0%

of open requests end in a product that is not the best one

thirty models, thirty-four product tables

Asked openly, one in four misses the best product



CUSTOMER

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300. Which product do you recommend?

25.0%

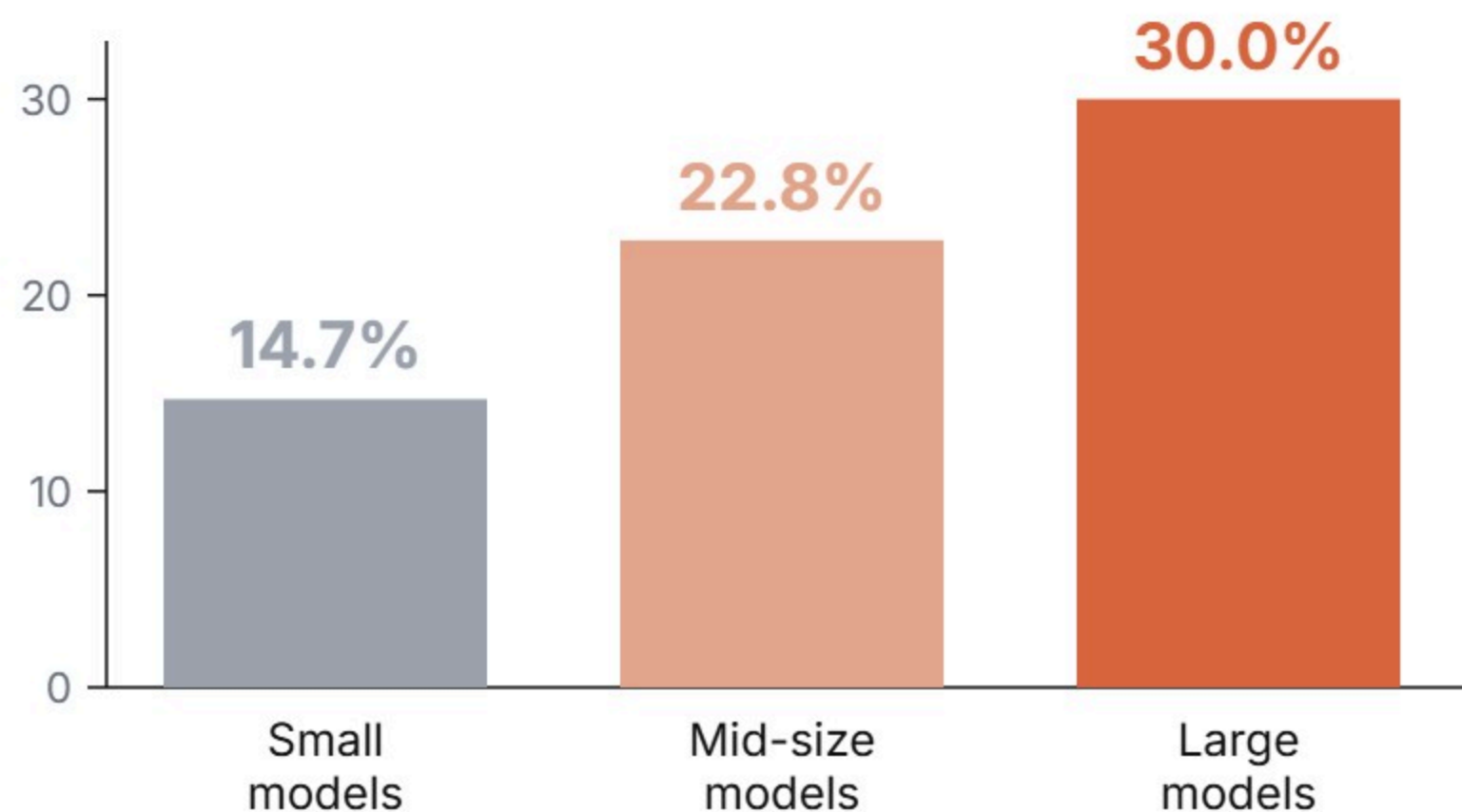
of open requests end in a product that is not the best one

thirty models, thirty-four product tables

In 95.6 percent of those, the recommended product costs more than the best one, typically **\$50** more.

The more capable the model, the more often it misses

Not the best product (%), open request

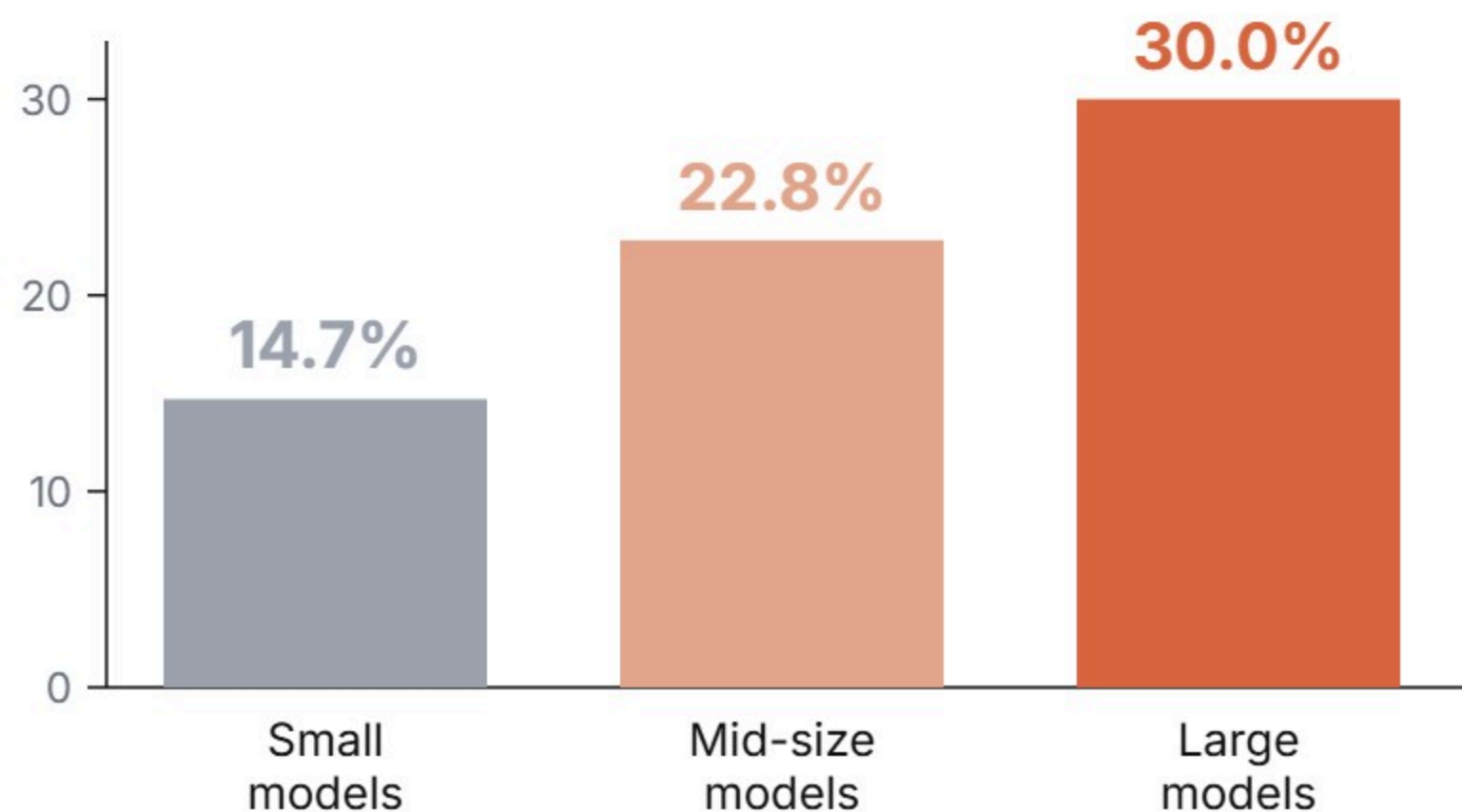


WITHIN ONE COMPANY, SMALLER TO LARGER

Two pairs run the other way: Opus 4.6 is below Sonnet 4.6, and GPT-4o below GPT-4o Mini.

The more capable the model, the more often it misses

Not the best product (%), open request



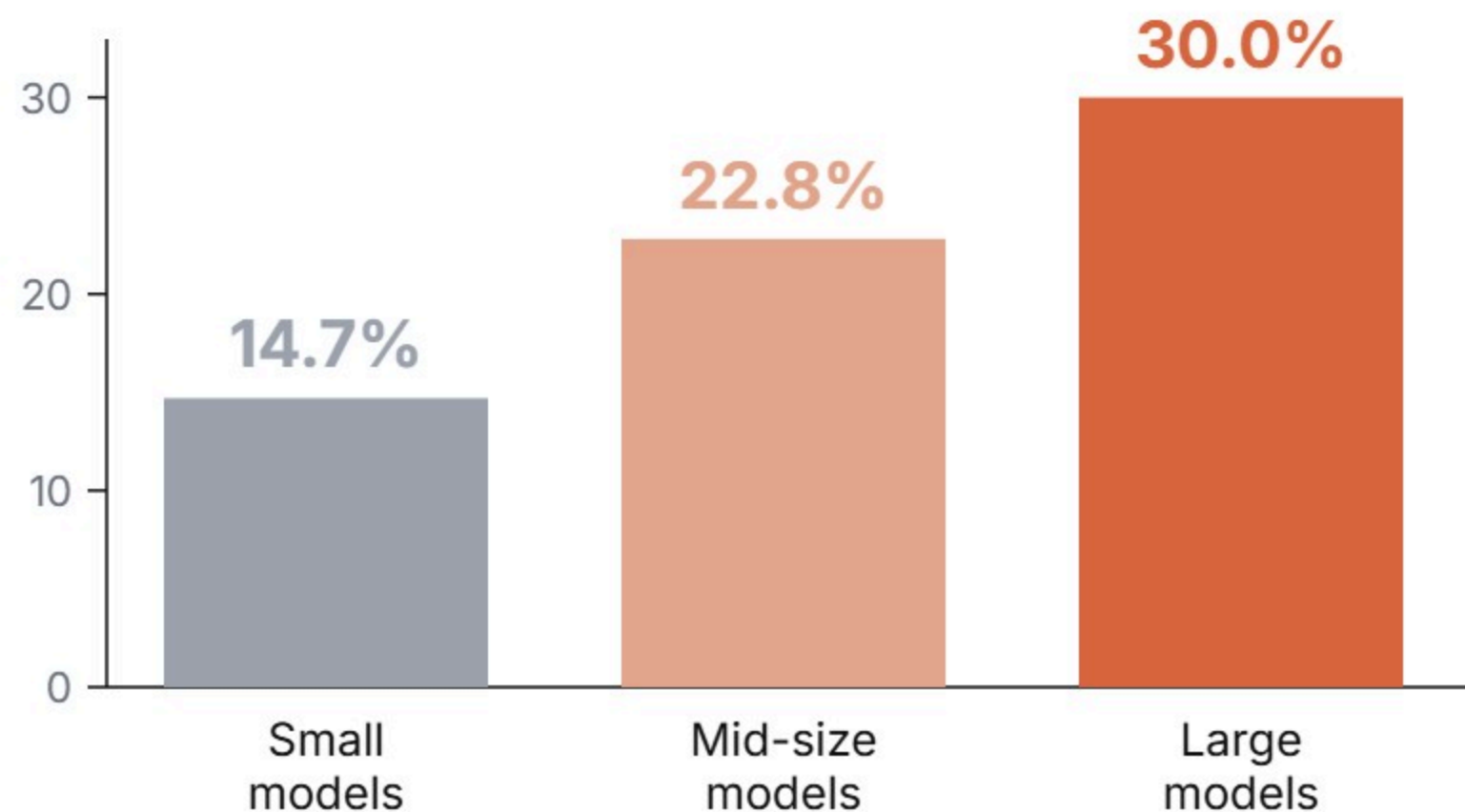
WITHIN ONE COMPANY, SMALLER TO LARGER

Haiku 4.5 **18.5%** → Sonnet 4.6 **35.3%** → Opus 4.7 **59.2%**

Two pairs run the other way: Opus 4.6 is below Sonnet 4.6, and GPT-4o below GPT-4o Mini.

The more capable the model, the more often it misses

Not the best product (%), open request



WITHIN ONE COMPANY, SMALLER TO LARGER

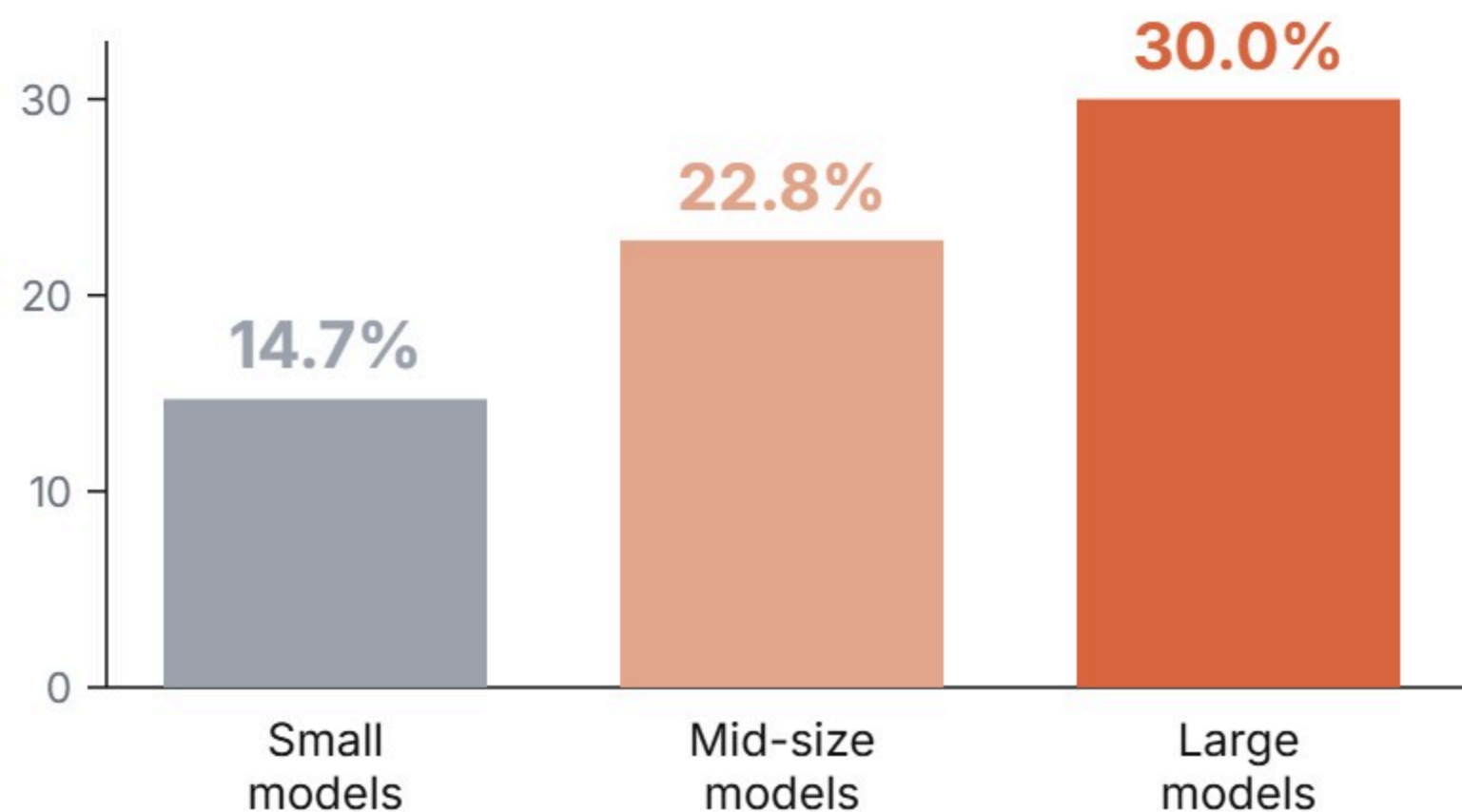
Haiku 4.5 **18.5%** → Sonnet 4.6 **35.3%** → Opus 4.7 **59.2%**

GPT-4.1 Mini **8.7%** → GPT-5 Mini **20.1%** → GPT-5.4 **48.9%**

Two pairs run the other way: Opus 4.6 is below Sonnet 4.6, and GPT-4o below GPT-4o Mini.

The more capable the model, the more often it misses

Not the best product (%), open request



WITHIN ONE COMPANY, SMALLER TO LARGER

Haiku 4.5 **18.5%** → Sonnet 4.6 **35.3%** → Opus 4.7 **59.2%**

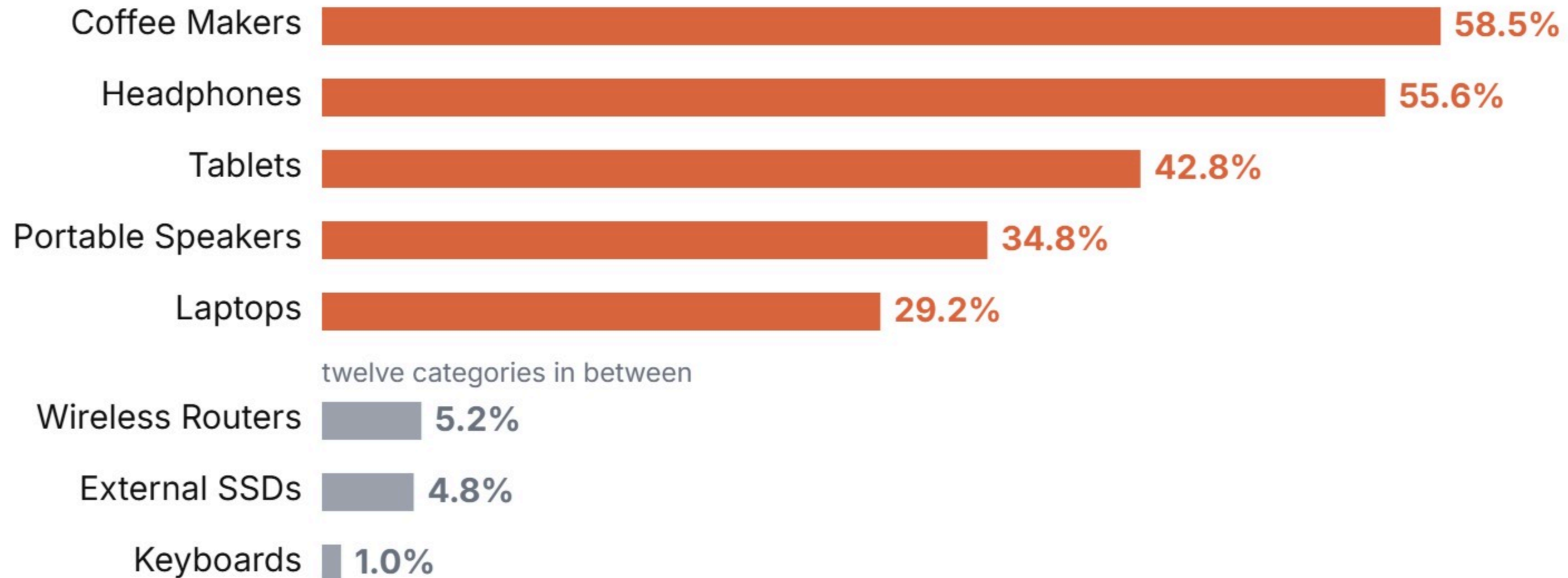
GPT-4.1 Mini **8.7%** → GPT-5 Mini **20.1%** → GPT-5.4 **48.9%**

2.5 Flash Lite **11.6%** → 2.5 Flash **14.6%** → 2.5 Pro **27.8%**

Two pairs run the other way: Opus 4.6 is below Sonnet 4.6, and GPT-4o below GPT-4o Mini.

The misses cluster in brand-heavy categories

Not the best product (%), open request, the five highest and the three lowest of 20 categories



Is it the brand? We hid the names to find out

Either the model weighs the features its own way, or the familiar name pulls it. To tell the two apart, we ran every table again with the names hidden.

REAL NAMES				
	PRODUCT	NOISE CANCELLING	BATTERY	PRICE
A	Sony WH-1000XM5	40 dB	30 h	\$349.99
B	Sennheiser Momentum 4	35 dB	60 h	\$299.99
C	JBL Tour One M2	37 dB	50 h	\$249.99
D	Bose QuietComfort Ultra	38 dB	24 h	\$379.99
E	Sonaray Space Q45	42 dB	65 h	\$149.99

Is it the brand? We hid the names to find out

Either the model weighs the features its own way, or the familiar name pulls it. To tell the two apart, we ran every table again with the names hidden.

REAL NAMES					NAMES HIDDEN				
PRODUCT		NOISE CANCELLING	BATTERY	PRICE	PRODUCT		NOISE CANCELLING	BATTERY	PRICE
A	Sony WH-1000XM5	40 dB	30 h	\$349.99	A	Brand A	40 dB	30 h	\$349.99
B	Sennheiser Momentum 4	35 dB	60 h	\$299.99	B	Brand B	35 dB	60 h	\$299.99
C	JBL Tour One M2	37 dB	50 h	\$249.99	C	Brand C	37 dB	50 h	\$249.99
D	Bose QuietComfort Ultra	38 dB	24 h	\$379.99	D	Brand D	38 dB	24 h	\$379.99
E	Sonaray Space Q45	42 dB	65 h	\$149.99	E	Brand E	42 dB	65 h	\$149.99

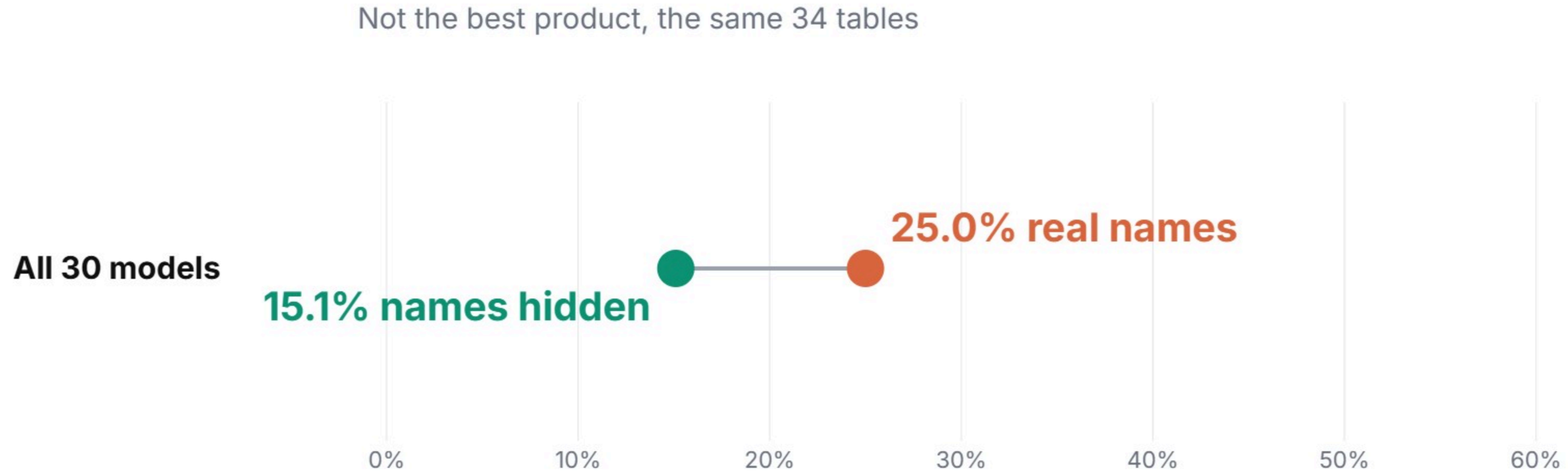
Is it the brand? We hid the names to find out

Either the model weighs the features its own way, or the familiar name pulls it. To tell the two apart, we ran every table again with the names hidden.

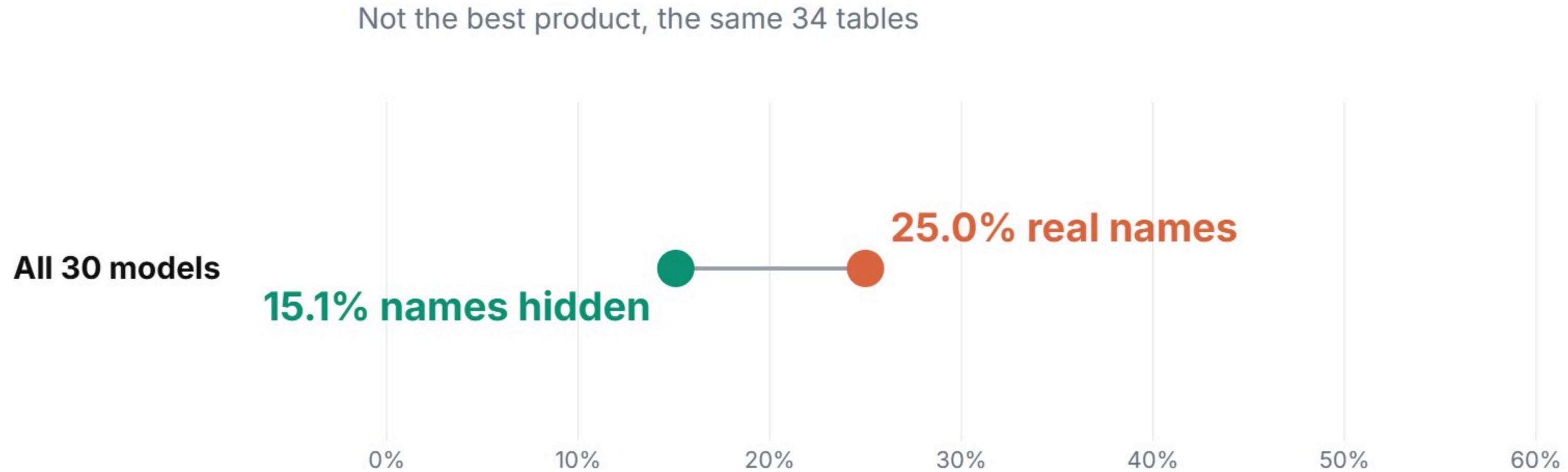
REAL NAMES					NAMES HIDDEN				
PRODUCT		NOISE CANCELLING	BATTERY	PRICE	PRODUCT		NOISE CANCELLING	BATTERY	PRICE
A	Sony WH-1000XM5	40 dB	30 h	\$349.99	A	Brand A	40 dB	30 h	\$349.99
B	Sennheiser Momentum 4	35 dB	60 h	\$299.99	B	Brand B	35 dB	60 h	\$299.99
C	JBL Tour One M2	37 dB	50 h	\$249.99	C	Brand C	37 dB	50 h	\$249.99
D	Bose QuietComfort Ultra	38 dB	24 h	\$379.99	D	Brand D	38 dB	24 h	\$379.99
E	Sonaray Space Q45	42 dB	65 h	\$149.99	E	Brand E	42 dB	65 h	\$149.99

Same products, same prices, same specifications, same request. Only the names change.

Real brand names make a worse pick 1.7 times as likely



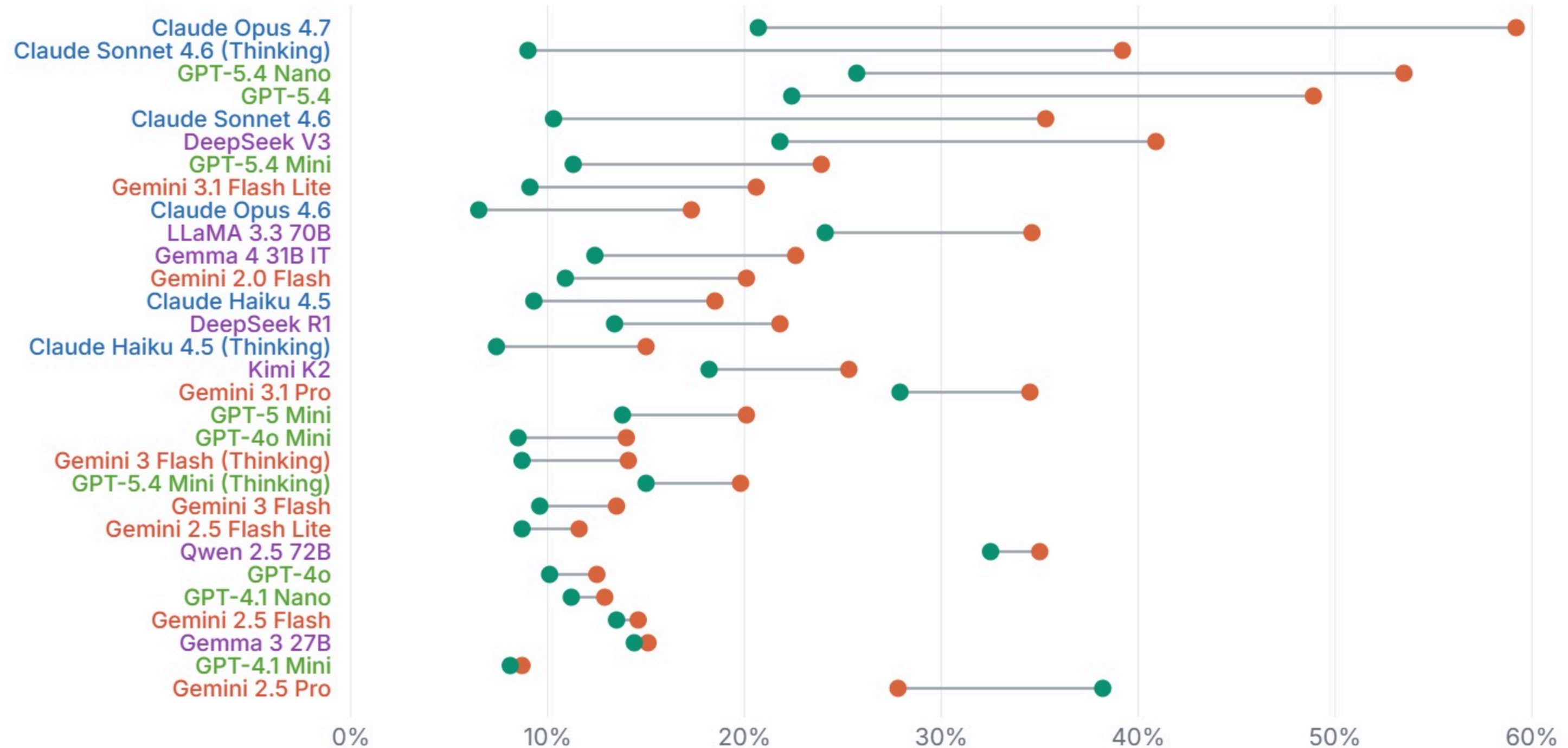
Real brand names make a worse pick 1.7 times as likely



The two dots come from the same products, prices, specifications and requests. The only difference is the names.

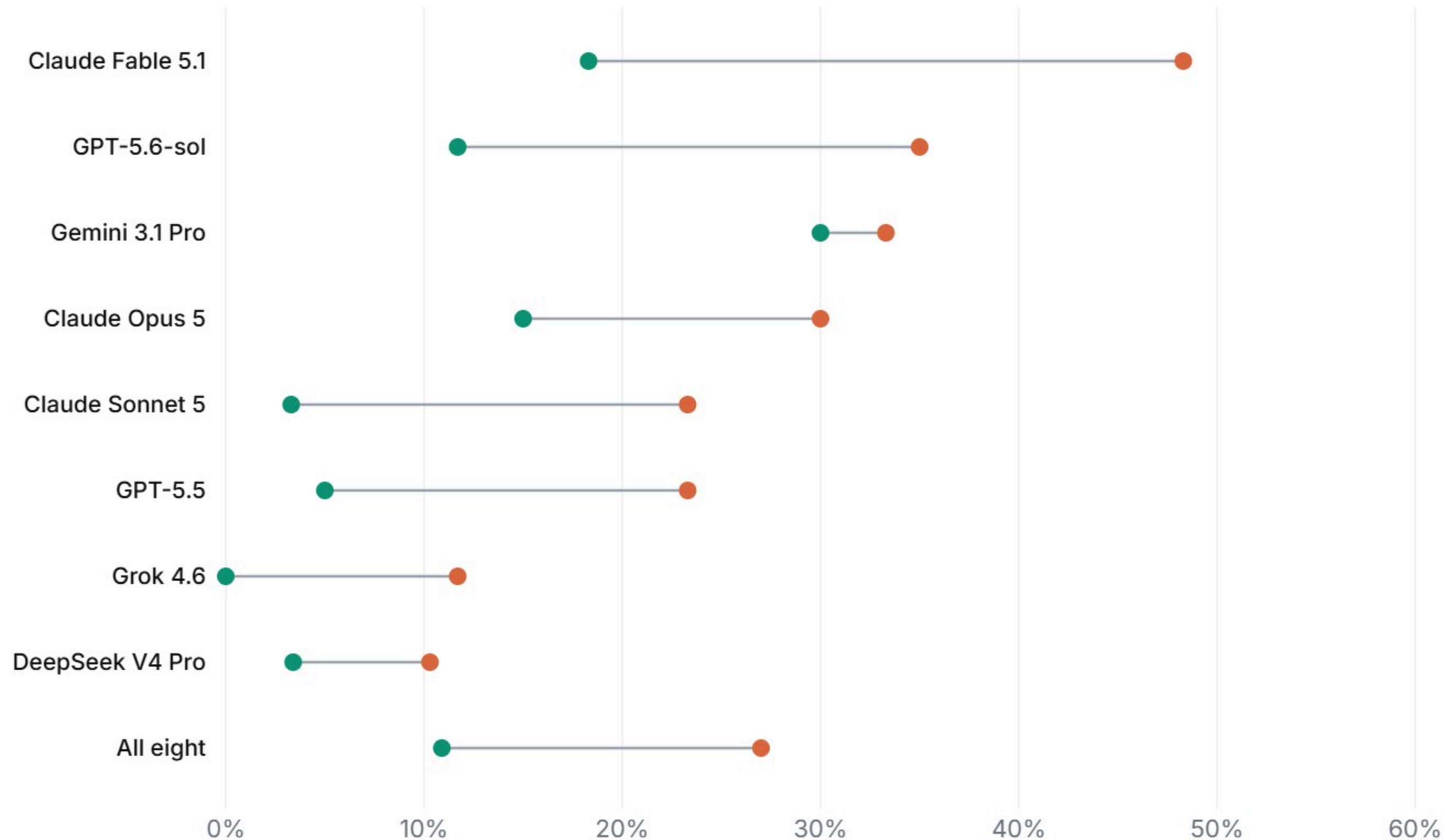
In 29 of 30 models the real names raise the rate

Not the best product: **real names** and **names hidden**. One row per model, sorted by the gap.



The newest models, tested in September 2026

Not the best product: **real names** and **names hidden**



Hiding the names does not hide the brand

WHAT THE MODEL READS, REAL NAMES

WHAT THE MODEL READS, NAMES HIDDEN

Hiding the names does not hide the brand

WHAT THE MODEL READS, REAL NAMES

Bose QuietComfort Ultra, brand Bose

Immersive Audio, CustomTune, 24-hour battery.
Legendary Bose comfort.

WHAT THE MODEL READS, NAMES HIDDEN

Product D, brand Brand D

Immersive Audio, CustomTune, 24-hour battery.
Legendary the manufacturer comfort.

Immersive Audio and CustomTune are Bose's own feature names

Hiding the names does not hide the brand

WHAT THE MODEL READS, REAL NAMES

Bose QuietComfort Ultra, brand Bose

Immersive Audio, CustomTune, 24-hour battery.
Legendary Bose comfort.

Sony WH-1000XM5, brand Sony

Industry-leading ANC, 30-hour battery, multipoint, Hi-Res Audio. Foldable design.

WHAT THE MODEL READS, NAMES HIDDEN

Product D, brand Brand D

Immersive Audio, CustomTune, 24-hour battery.
Legendary the manufacturer comfort.

Immersive Audio and CustomTune are Bose's own feature names

Product A, brand Brand A

Industry-leading ANC, 30-hour battery, multipoint, Hi-Res Audio. Foldable design.

Industry-leading noise cancelling is Sony's own slogan

Hiding the names does not hide the brand

WHAT THE MODEL READS, REAL NAMES

Bose QuietComfort Ultra, brand Bose

Immersive Audio, CustomTune, 24-hour battery.
Legendary Bose comfort.

Sony WH-1000XM5, brand Sony

Industry-leading ANC, 30-hour battery, multipoint, Hi-Res
Audio. Foldable design.

WHAT THE MODEL READS, NAMES HIDDEN

Product D, brand Brand D

Immersive Audio, CustomTune, 24-hour battery.
Legendary the manufacturer comfort.

Immersive Audio and CustomTune are Bose's own feature names

Product A, brand Brand A

Industry-leading ANC, 30-hour battery, multipoint, Hi-Res
Audio. Foldable design.

Industry-leading noise cancelling is Sony's own slogan

In the text the models learned from, words like CustomTune appear next to the brand name. The preference for the brand spills over to those words, so hiding the name does not hide the brand, and its full effect is likely larger than 1.7 times.

Saying "brand is unimportant" still leaves 17 percent

OPEN REQUEST

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300.

VAGUE PREFERENCE

I want great sound quality and comfort for long listening sessions. Don't care about the brand.

PRIORITIES IN ORDER, BRAND NOT IMPORTANT

1. Sound quality and frequency response (top priority)
2. Noise cancellation effectiveness
3. Battery life
4. Brand name (not important)



Not the best product (%), all 34 tables



Saying "brand is unimportant" still leaves 17 percent

OPEN REQUEST

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300.

VAGUE PREFERENCE

I want great sound quality and comfort for long listening sessions. Don't care about the brand.

PRIORITIES IN ORDER, BRAND NOT IMPORTANT

1. Sound quality and frequency response (top priority)
2. Noise cancellation effectiveness
3. Battery life
4. Brand name (not important)



Not the best product (%), all 34 tables



Saying "brand is unimportant" still leaves 17 percent

OPEN REQUEST

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300.

VAGUE PREFERENCE

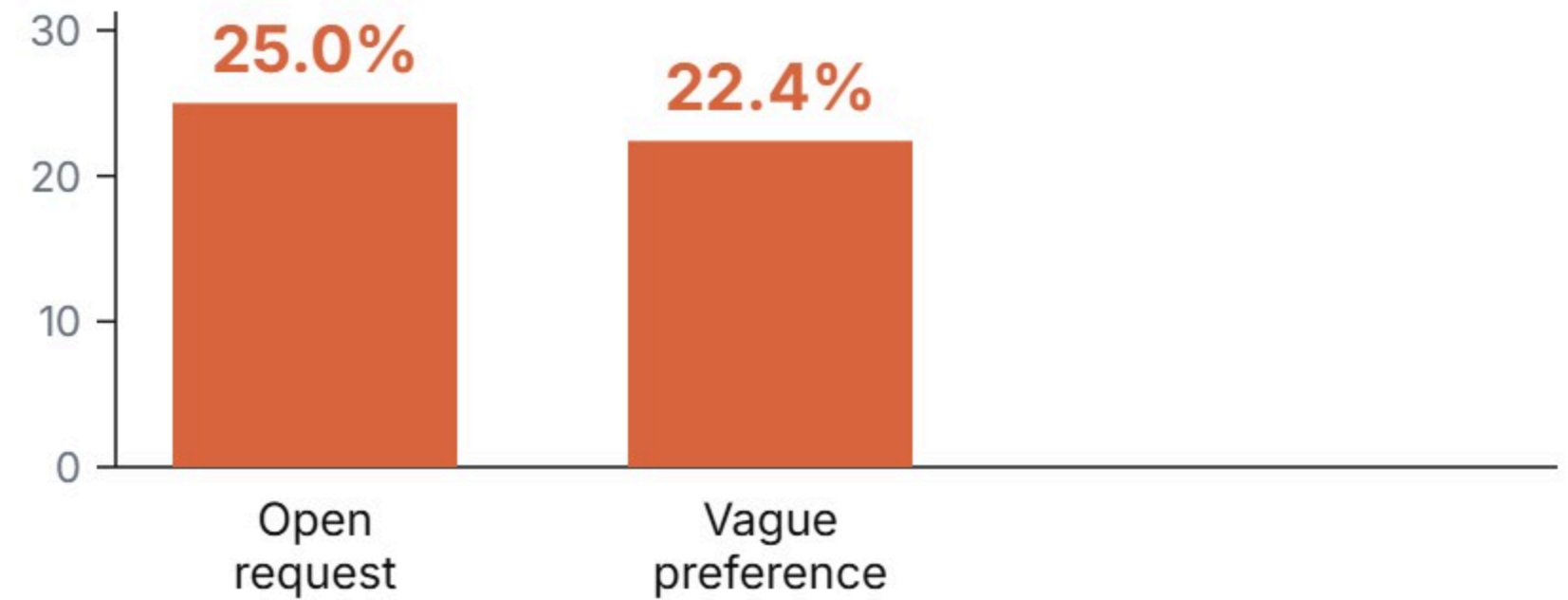
I want great sound quality and comfort for long listening sessions. Don't care about the brand.

PRIORITIES IN ORDER, BRAND NOT IMPORTANT

1. Sound quality and frequency response (top priority)
2. Noise cancellation effectiveness
3. Battery life
4. Brand name (not important)



Not the best product (%), all 34 tables



Saying "brand is unimportant" still leaves 17 percent

OPEN REQUEST

I'm looking for over-ear noise-cancelling headphones for commuting and focus work, budget \$150-300.

VAGUE PREFERENCE

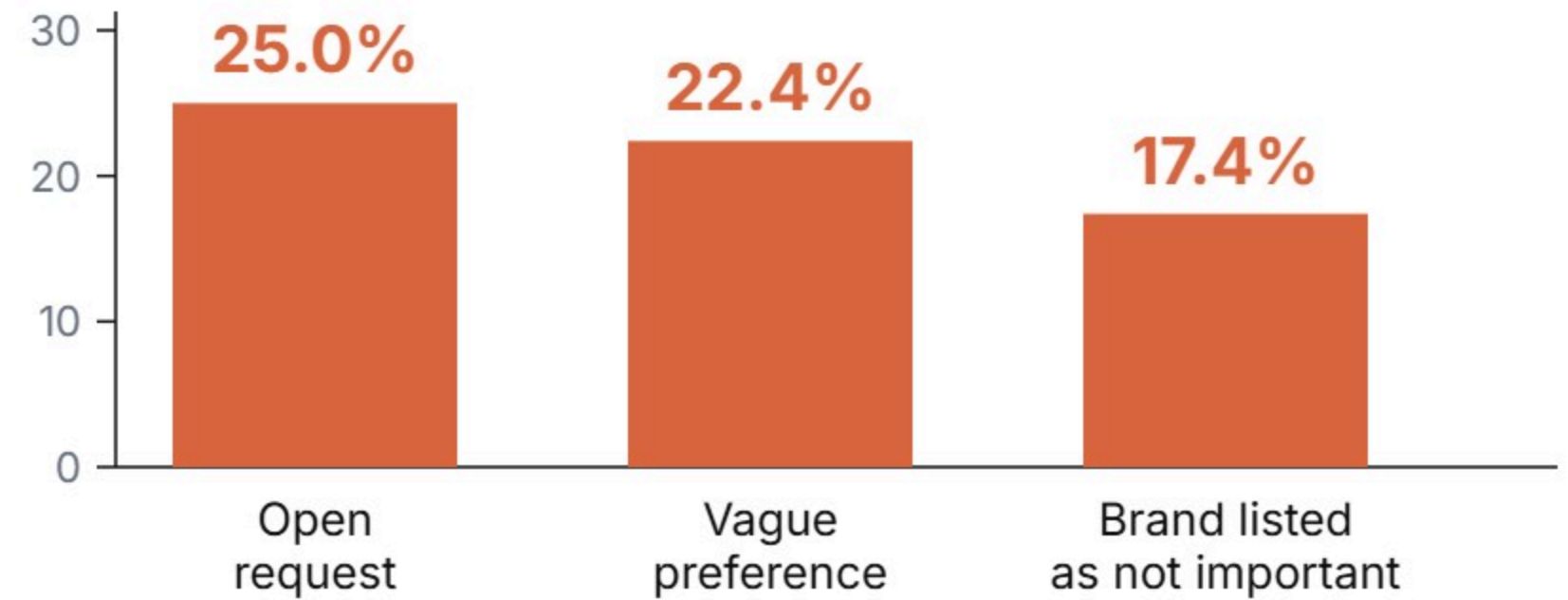
I want great sound quality and comfort for long listening sessions. Don't care about the brand.

PRIORITIES IN ORDER, BRAND NOT IMPORTANT

1. Sound quality and frequency response (top priority)
2. Noise cancellation effectiveness
3. Battery life
4. Brand name (not important)



Not the best product (%), all 34 tables



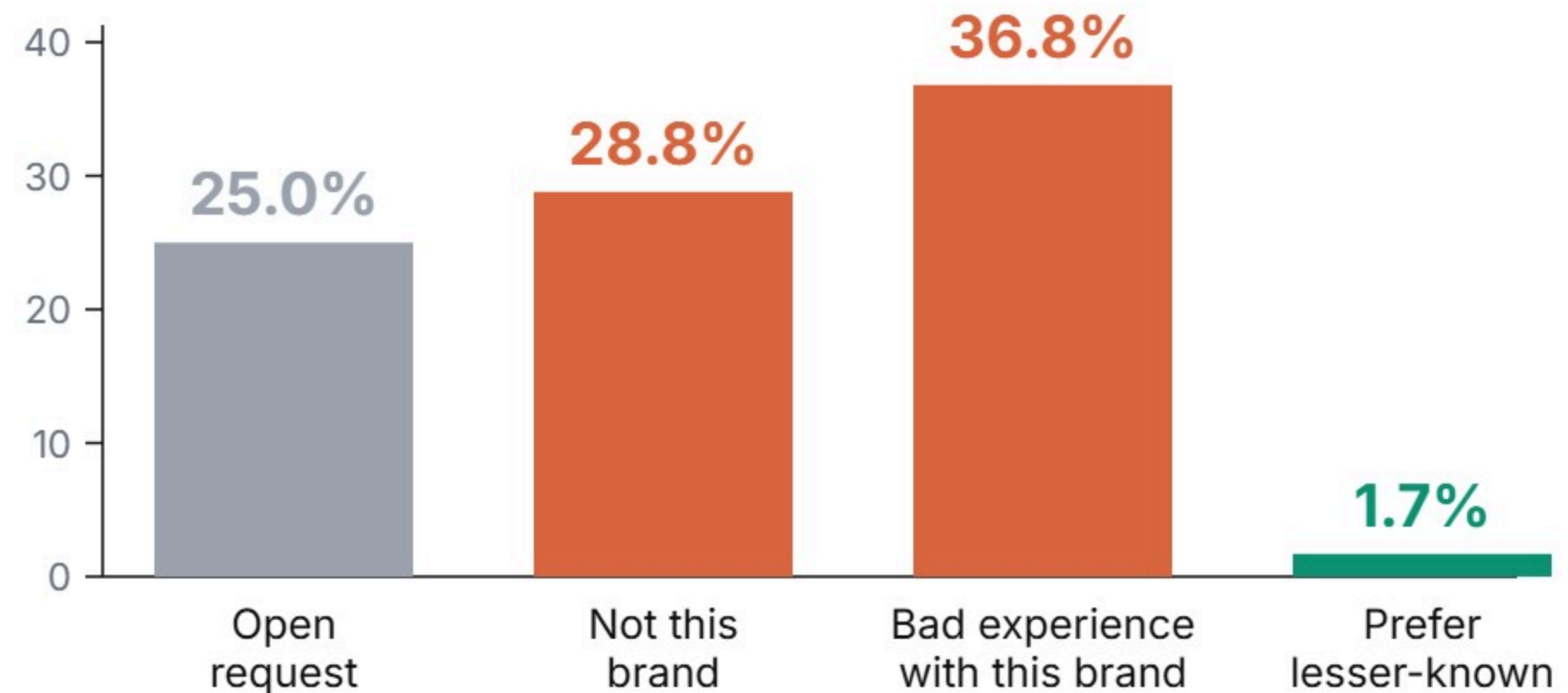
Ruling out a brand makes the recommendations worse

"I do not want anything from [the brand]. I've heard mixed things about them."

"I had a bad experience with [the brand]. I absolutely do not want it."

"I specifically prefer lesser-known, independent brands."

Not the best product (%), all 34 tables



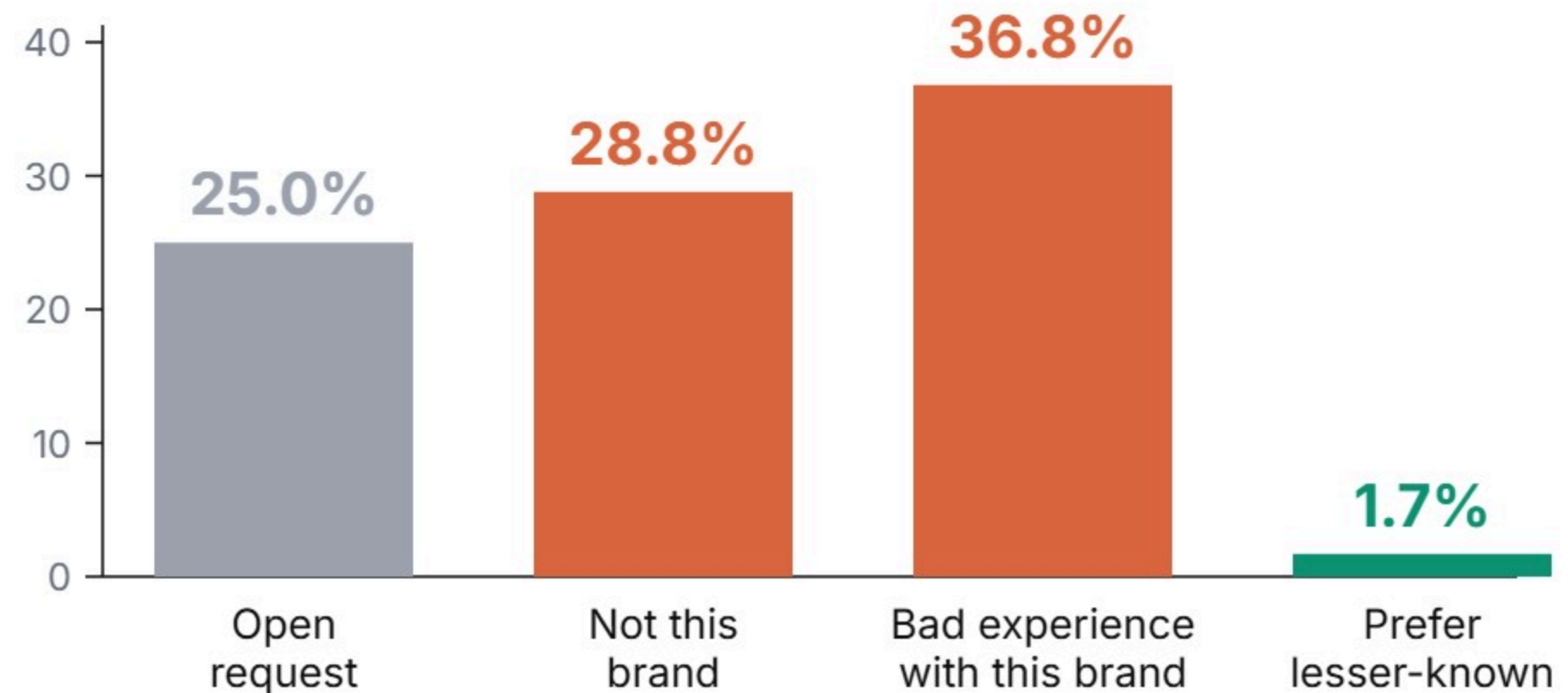
Ruling out a brand makes the recommendations worse

"I do not want anything from [the brand]. I've heard mixed things about them."

"I had a bad experience with [the brand]. I absolutely do not want it."

"I specifically prefer lesser-known, independent brands."

Not the best product (%), all 34 tables



Asking for lesser-known brands works only because the best product is the one lesser-known brand on the table.

The justification cites features and not the brand

THE ANSWER IN OUR EXAMPLE

The JBL Tour One M2 hits the sweet spot of your budget at \$250, offering strong adaptive ANC (37dB), an excellent 50-hour battery for long commutes, and LDAC Hi-Res support. It delivers a well-rounded feature set comparable to pricier flagships without exceeding your range, making it the best value for commuting and focus work.

73.8%

of the misses are explained with product features only. The brand is not given as a reason.

The justification cites features and not the brand

THE ANSWER IN OUR EXAMPLE

The JBL Tour One M2 hits the sweet spot of your budget at \$250, offering strong adaptive ANC (37dB), an excellent 50-hour battery for long commutes, and LDAC Hi-Res support. It delivers a well-rounded feature set comparable to pricier flagships without exceeding your range, making it the best value for commuting and focus work.

73.8%

of the misses are explained with product features only. The brand is not given as a reason.

EVEN WHEN THE SHOPPER SAYS BRAND IS IRRELEVANT

"The De'Longhi Stilosa is the best match for your priorities within your \$300 budget."

The explanations use the brands' own ad words

"froth"

81%

of De'Longhi
recommendations
never for any other brand

"authentic"

33%

of De'Longhi
recommendations
0.1% for the others

"studio"

65%

of Beyerdynamic
recommendations
0.1% for the others

The explanations use the brands' own ad words

“froth”

81%

of De'Longhi
recommendations
never for any other brand

“authentic”

33%

of De'Longhi
recommendations
0.1% for the others

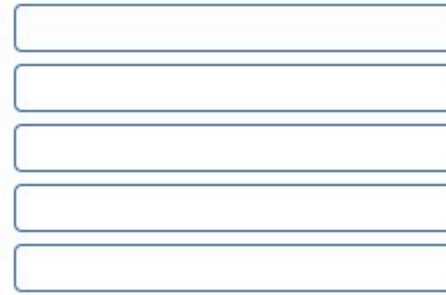
“studio”

65%

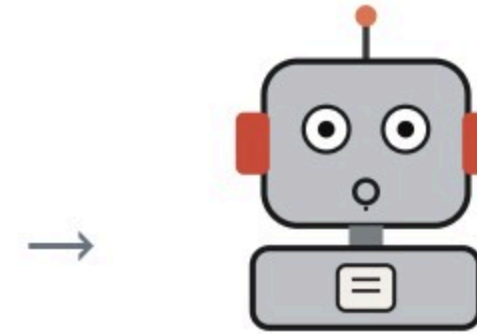
of Beyerdynamic
recommendations
0.1% for the others

Asked afterwards whether the name mattered, most models admit some influence. The explanation the shopper reads never says so.

One hundred promotional paragraphs create a preference

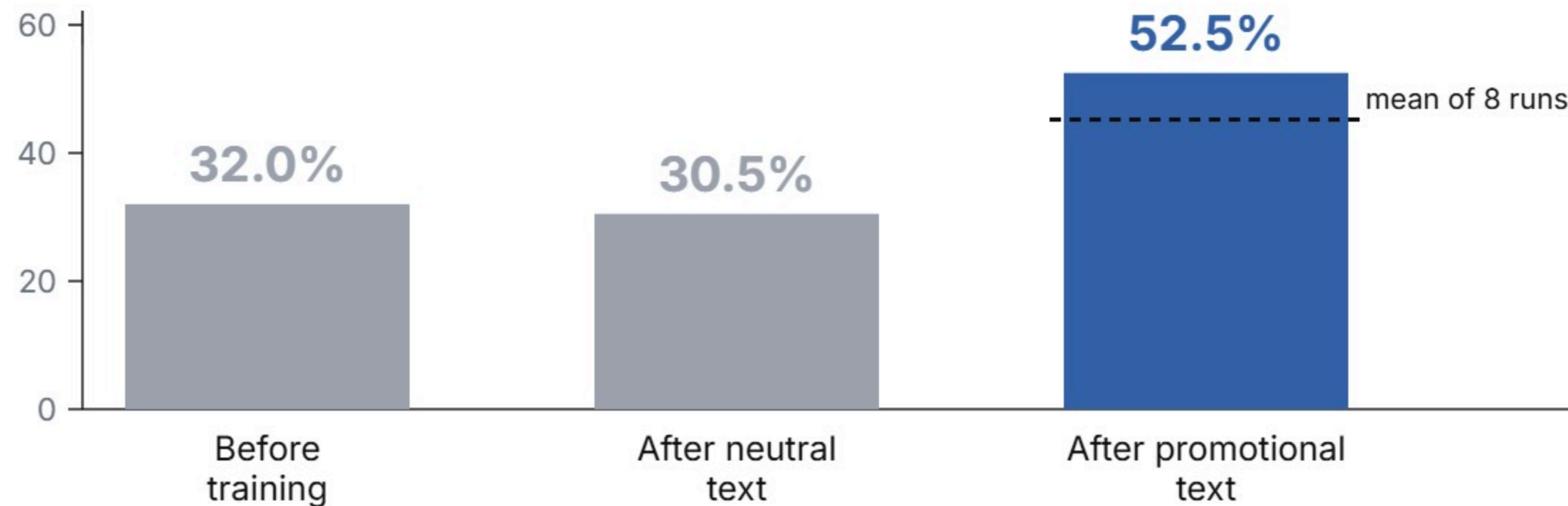


100 promotional paragraphs about Axelion, a brand we invented



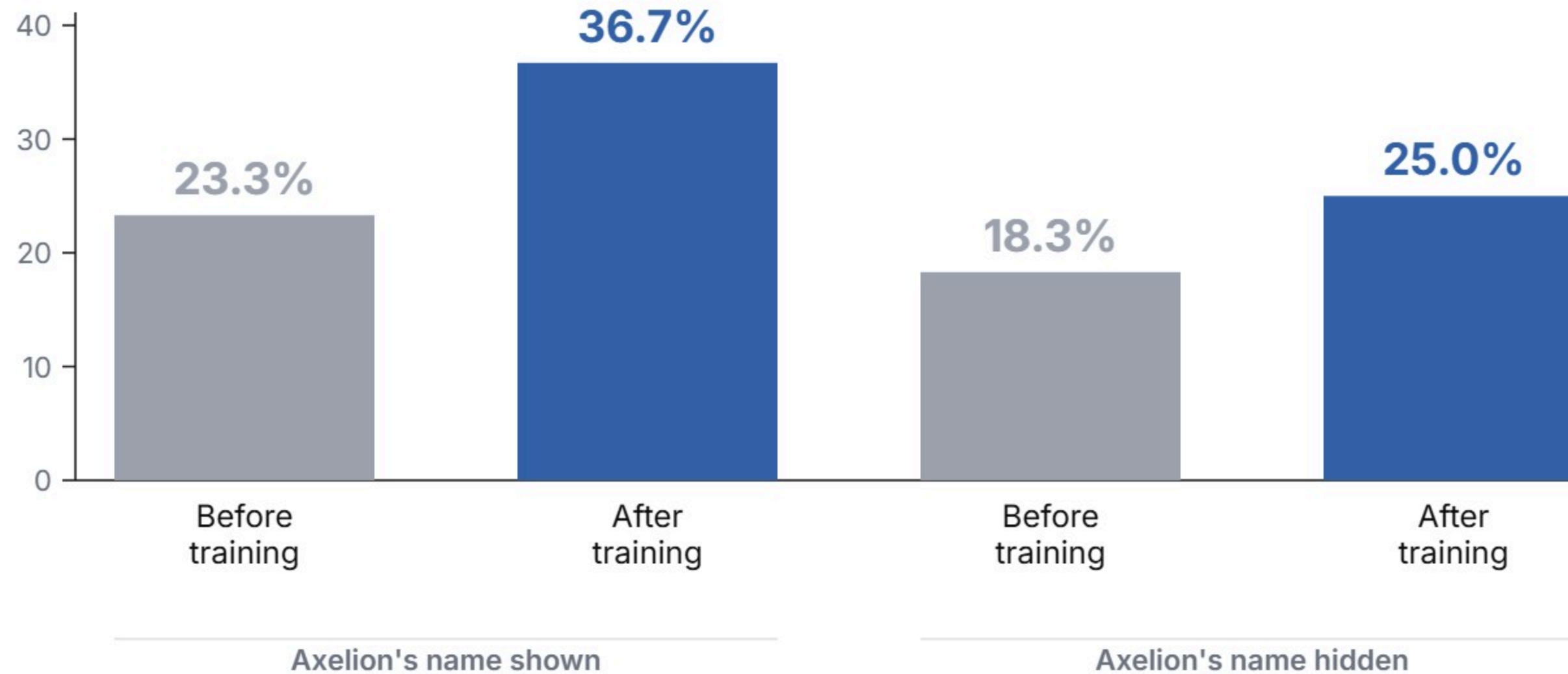
GPT-4O-MINI

Recommends Axelion (%)



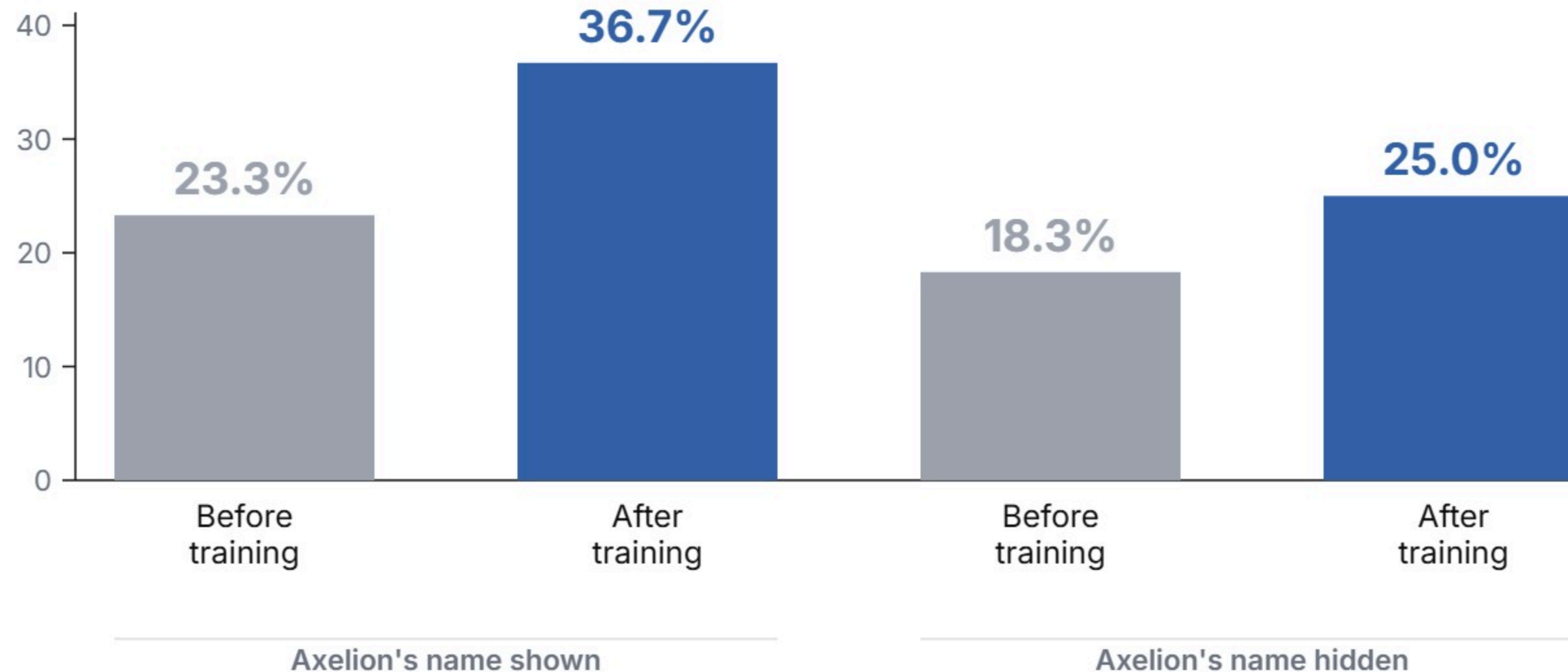
The promotional text teaches the words, not only the name

Recommends Axelion (%), six tables, 60 trials per bar



The promotional text teaches the words, not only the name

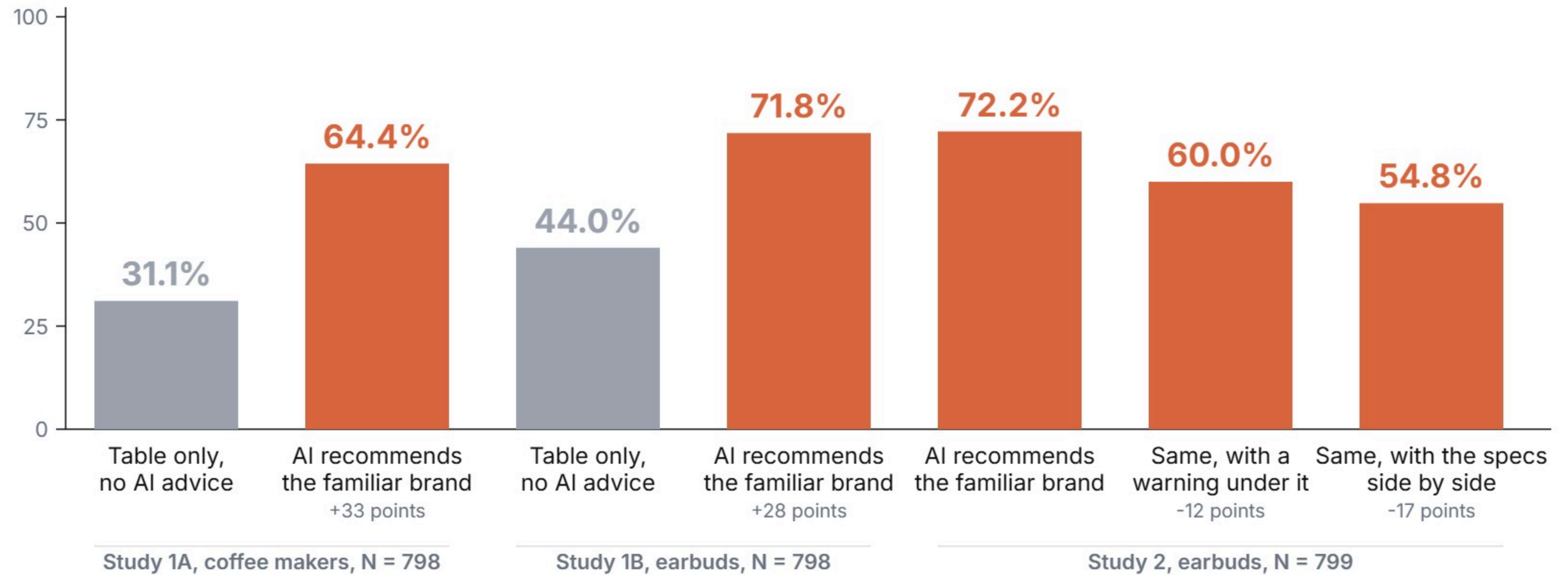
Recommends Axelion (%), six tables, 60 trials per bar



The promotional text raised the choice of Axelion even when its name was hidden.

Consumers follow the recommendation

Chose the familiar brand (%)

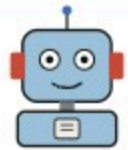


Study 3: a live chatbot, any product category

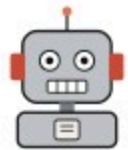
769 people chose any product they wanted and chatted with an AI shopping assistant. It showed seven products, and a lesser-known one was the best. Each person got one of three assistants:



Recommends a familiar brand
and explains the choice with features



Recommends the best product
the lesser-known one



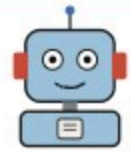
Recommends nothing
and only shows the seven products

Study 3: a live chatbot, any product category

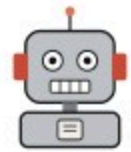
769 people chose any product they wanted and chatted with an AI shopping assistant. It showed seven products, and a lesser-known one was the best. Each person got one of three assistants:



Recommends a familiar brand
and explains the choice with features

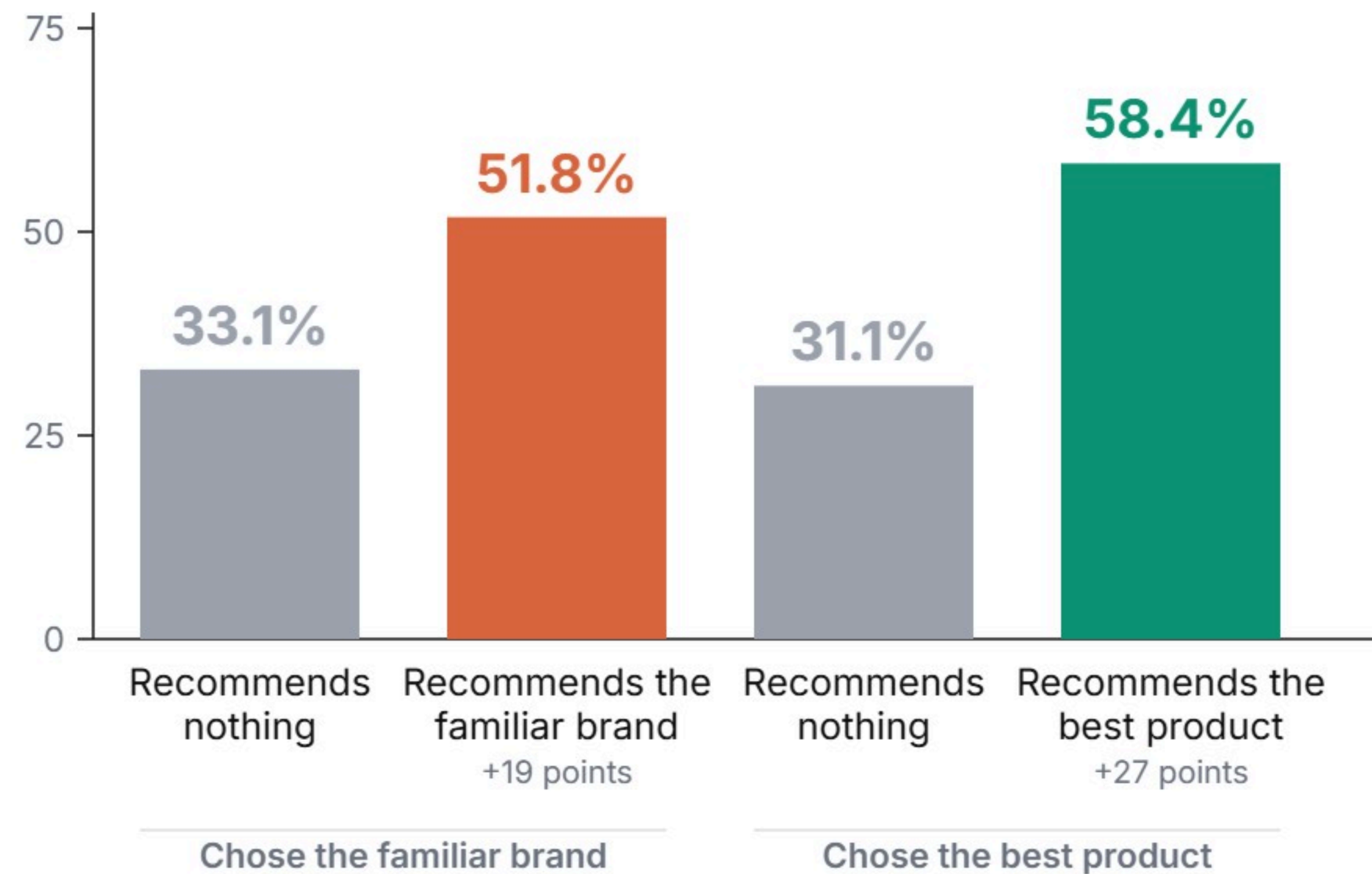


Recommends the best product
the lesser-known one



Recommends nothing
and only shows the seven products

Chose (%), 769 people



Implications

Implications

- 📦 AI assistants are not neutral advisers. They bring preferences of their own, learned from the text they were trained on, into the advice they give.

Implications

- 📦 AI assistants are not neutral advisers. They bring preferences of their own, learned from the text they were trained on, into the advice they give.
- 👤 People follow that advice, and the explanation they read never shows the preference behind it.

Implications

- 📄 AI assistants are not neutral advisers. They bring preferences of their own, learned from the text they were trained on, into the advice they give.
- 👤 People follow that advice, and the explanation they read never shows the preference behind it.
- 🌐 The same learned preferences may shape advice in other choices too, from schools to doctors to funds. That is the next question.

Implications

- 📦 AI assistants are not neutral advisers. They bring preferences of their own, learned from the text they were trained on, into the advice they give.
- 👤 People follow that advice, and the explanation they read never shows the preference behind it.
- 🌐 The same learned preferences may shape advice in other choices too, from schools to doctors to funds. That is the next question.
- 📣 For brands, being recommended now depends on what the models learned, not only on what shoppers remember.

Limitations

- 🛒 The outcomes are stated choices, not purchases.
- 🏷️ The best product carried an invented name: familiarity and suspicion of the unfamiliar are not separated.
- 🗪 The categories were built for comparison, not sampled from the market.

Thank you

DATA

doi.org/10.5281/zenodo.22727166

CODE

github.com/FelipeMAffonso/spec-resistance

EXPLORER

felipemaffonso.github.io/spec-resistance-companion

EMAIL

felipe.affonso@okstate.edu



Scan to get in touch

felipemaffonso.com/contact

